

Supplement: Dispatch Cost and Mechanism Activation

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1 Complete Difference and Equivalence Family

Gain is the mean of seed-level relative effects. Difference intervals and tests share a Student approximation and a joint Bonferroni family containing both difference and TOST hypotheses. The equivalence margin is one percentage point. Non-significance, equivalence, and exact sampled identity are distinct.

Cell	Candidate	Comparator	Mean gain (%)	Simultaneous CI (%)	Difference p	TOST p
continuous RSSI	Proposed	Compute-shrink	1.75	[1.43, 2.07]	< 0.0001	1.0000
continuous RSSI	Proposed	Proposed-allPrefetch	0.00	[0.00, 0.00]	1.0000	< 0.0001
continuous RSSI	Proposed	Proposed-noCutoff	0.12	[-0.01, 0.24]	0.1245	< 0.0001
continuous RSSI	Proposed	Proposed-noPrefetch	0.41	[0.22, 0.60]	< 0.0001	< 0.0001
continuous RSSI	Proposed	Proposed-noRSSI	0.04	[-0.09, 0.18]	1.0000	< 0.0001
continuous RSSI	Proposed	Proposed-noRTT	0.00	[0.00, 0.00]	1.0000	< 0.0001
continuous RSSI	Proposed	Tuned-fixed	0.46	[0.33, 0.58]	< 0.0001	< 0.0001
churn	Proposed	Compute-shrink	2.42	[1.84, 3.00]	< 0.0001	1.0000
churn	Proposed	Proposed-allPrefetch	0.16	[-0.00, 0.32]	0.0691	< 0.0001
churn	Proposed	Proposed-noCutoff	-0.11	[-0.27, 0.04]	1.0000	< 0.0001
churn	Proposed	Proposed-noPrefetch	2.42	[1.84, 3.00]	< 0.0001	1.0000
churn	Proposed	Proposed-noRSSI	0.00	[0.00, 0.00]	1.0000	< 0.0001
churn	Proposed	Proposed-noRTT	0.00	[0.00, 0.00]	1.0000	< 0.0001
churn	Proposed	Tuned-fixed	-1.52	[-1.81, -1.23]	< 0.0001	1.0000
combined	Proposed	Compute-shrink	10.12	[9.19, 11.05]	< 0.0001	1.0000
combined	Proposed	Proposed-allPrefetch	0.12	[-0.29, 0.52]	1.0000	< 0.0001
combined	Proposed	Proposed-noCutoff	-0.13	[-0.30, 0.05]	1.0000	< 0.0001
combined	Proposed	Proposed-noPrefetch	0.96	[0.38, 1.54]	< 0.0001	1.0000
combined	Proposed	Proposed-noRSSI	9.52	[8.68, 10.36]	< 0.0001	1.0000
combined	Proposed	Proposed-noRTT	0.00	[0.00, 0.00]	1.0000	< 0.0001
combined	Proposed	Tuned-fixed	-4.65	[-5.34, -3.97]	< 0.0001	1.0000
uniform	Proposed	Compute-shrink	1.45	[1.14, 1.75]	< 0.0001	1.0000
uniform	Proposed	Proposed-allPrefetch	0.00	[0.00, 0.00]	1.0000	< 0.0001
uniform	Proposed	Proposed-noCutoff	0.11	[-0.03, 0.26]	0.8254	< 0.0001
uniform	Proposed	Proposed-noPrefetch	1.45	[1.14, 1.75]	< 0.0001	1.0000
uniform	Proposed	Proposed-noRSSI	0.00	[0.00, 0.00]	1.0000	< 0.0001
uniform	Proposed	Proposed-noRTT	0.00	[0.00, 0.00]	1.0000	< 0.0001
uniform	Proposed	Tuned-fixed	0.08	[-0.06, 0.22]	1.0000	< 0.0001
weak links	Proposed	Compute-shrink	9.34	[8.56, 10.11]	< 0.0001	1.0000
weak links	Proposed	Proposed-allPrefetch	0.00	[0.00, 0.00]	1.0000	< 0.0001
weak links	Proposed	Proposed-noCutoff	-0.07	[-0.24, 0.10]	1.0000	< 0.0001
weak links	Proposed	Proposed-noPrefetch	0.33	[-0.11, 0.77]	1.0000	< 0.0001
weak links	Proposed	Proposed-noRSSI	9.20	[8.50, 9.90]	< 0.0001	1.0000
weak links	Proposed	Proposed-noRTT	0.00	[0.00, 0.00]	1.0000	< 0.0001
weak links	Proposed	Tuned-fixed	-3.67	[-4.15, -3.19]	< 0.0001	1.0000
dispatch 0	Proposed	Compute-shrink	1.33	[0.98, 1.67]	< 0.0001	1.0000
dispatch 0	Proposed	Fixed-one	0.25	[0.08, 0.43]	< 0.0001	< 0.0001

Cell	Candidate	Comparator	Mean gain (%)	Simultaneous CI (%)	Difference p	TOST p
dispatch 0	Proposed	Tuned-fixed	0.25	[0.08, 0.43]	< 0.0001	< 0.0001
dispatch 1	Proposed	Compute-shrink	1.34	[0.99, 1.69]	< 0.0001	1.0000
dispatch 1	Proposed	Fixed-one	0.26	[0.08, 0.44]	< 0.0001	< 0.0001
dispatch 1	Proposed	Tuned-fixed	0.26	[0.08, 0.44]	< 0.0001	< 0.0001
dispatch 10	Proposed	Compute-shrink	1.32	[0.96, 1.69]	< 0.0001	1.0000
dispatch 10	Proposed	Fixed-one	0.26	[0.11, 0.42]	< 0.0001	< 0.0001
dispatch 10	Proposed	Tuned-fixed	0.26	[0.11, 0.42]	< 0.0001	< 0.0001
dispatch 100	Proposed	Compute-shrink	1.37	[1.00, 1.74]	< 0.0001	1.0000
dispatch 100	Proposed	Fixed-one	0.32	[0.16, 0.48]	< 0.0001	< 0.0001
dispatch 100	Proposed	Tuned-fixed	0.32	[0.16, 0.48]	< 0.0001	< 0.0001
dispatch 1000	Proposed	Compute-shrink	1.77	[1.37, 2.17]	< 0.0001	1.0000
dispatch 1000	Proposed	Fixed-one	0.39	[0.23, 0.56]	< 0.0001	< 0.0001
dispatch 1000	Proposed	Tuned-fixed	0.39	[0.23, 0.56]	< 0.0001	< 0.0001
dispatch 20000	Proposed	Compute-shrink	2.00	[1.63, 2.36]	< 0.0001	1.0000
dispatch 20000	Proposed	Fixed-one	64.94	[64.82, 65.05]	< 0.0001	1.0000
dispatch 20000	Proposed	Tuned-fixed	-32.10	[-32.74, -31.45]	< 0.0001	1.0000
dispatch 5000	Proposed	Compute-shrink	2.29	[1.83, 2.75]	< 0.0001	1.0000
dispatch 5000	Proposed	Fixed-one	3.81	[3.48, 4.15]	< 0.0001	1.0000
dispatch 5000	Proposed	Tuned-fixed	1.26	[1.01, 1.52]	< 0.0001	1.0000
independent RTT	Proposed	Compute-shrink	1.35	[1.05, 1.65]	< 0.0001	1.0000
independent RTT	Proposed	Proposed-allPrefetch	0.00	[0.00, 0.00]	1.0000	< 0.0001
independent RTT	Proposed	Proposed-noCutoff	0.15	[0.02, 0.28]	0.0050	< 0.0001
independent RTT	Proposed	Proposed-noPrefetch	0.41	[0.25, 0.58]	< 0.0001	< 0.0001
independent RTT	Proposed	Proposed-noRSSI	0.00	[0.00, 0.00]	1.0000	< 0.0001
independent RTT	Proposed	Proposed-noRTT	-0.12	[-0.26, 0.02]	0.2211	< 0.0001
independent RTT	Proposed	Tuned-fixed	0.21	[0.09, 0.34]	< 0.0001	< 0.0001
prefetch departure	Proposed	Compute-shrink	-3.12	[-4.77, -1.48]	< 0.0001	1.0000
prefetch departure	Proposed	Proposed-allPrefetch	0.59	[-0.54, 1.72]	1.0000	1.0000
prefetch departure	Proposed	Proposed-noCutoff	1.33	[0.87, 1.80]	< 0.0001	1.0000
prefetch departure	Proposed	Proposed-noPrefetch	-3.11	[-4.74, -1.48]	< 0.0001	1.0000
prefetch departure	Proposed	Proposed-noRSSI	0.07	[-0.12, 0.27]	1.0000	< 0.0001
prefetch departure	Proposed	Proposed-noRTT	0.00	[0.00, 0.00]	1.0000	< 0.0001
prefetch departure	Proposed	Tuned-fixed	-3.08	[-4.54, -1.62]	< 0.0001	1.0000
dispatch 0	Tuned-fixed	Fixed-one	0.00	[0.00, 0.00]	1.0000	< 0.0001
dispatch 1	Tuned-fixed	Fixed-one	0.00	[0.00, 0.00]	1.0000	< 0.0001
dispatch 10	Tuned-fixed	Fixed-one	0.00	[0.00, 0.00]	1.0000	< 0.0001
dispatch 100	Tuned-fixed	Fixed-one	0.00	[0.00, 0.00]	1.0000	< 0.0001

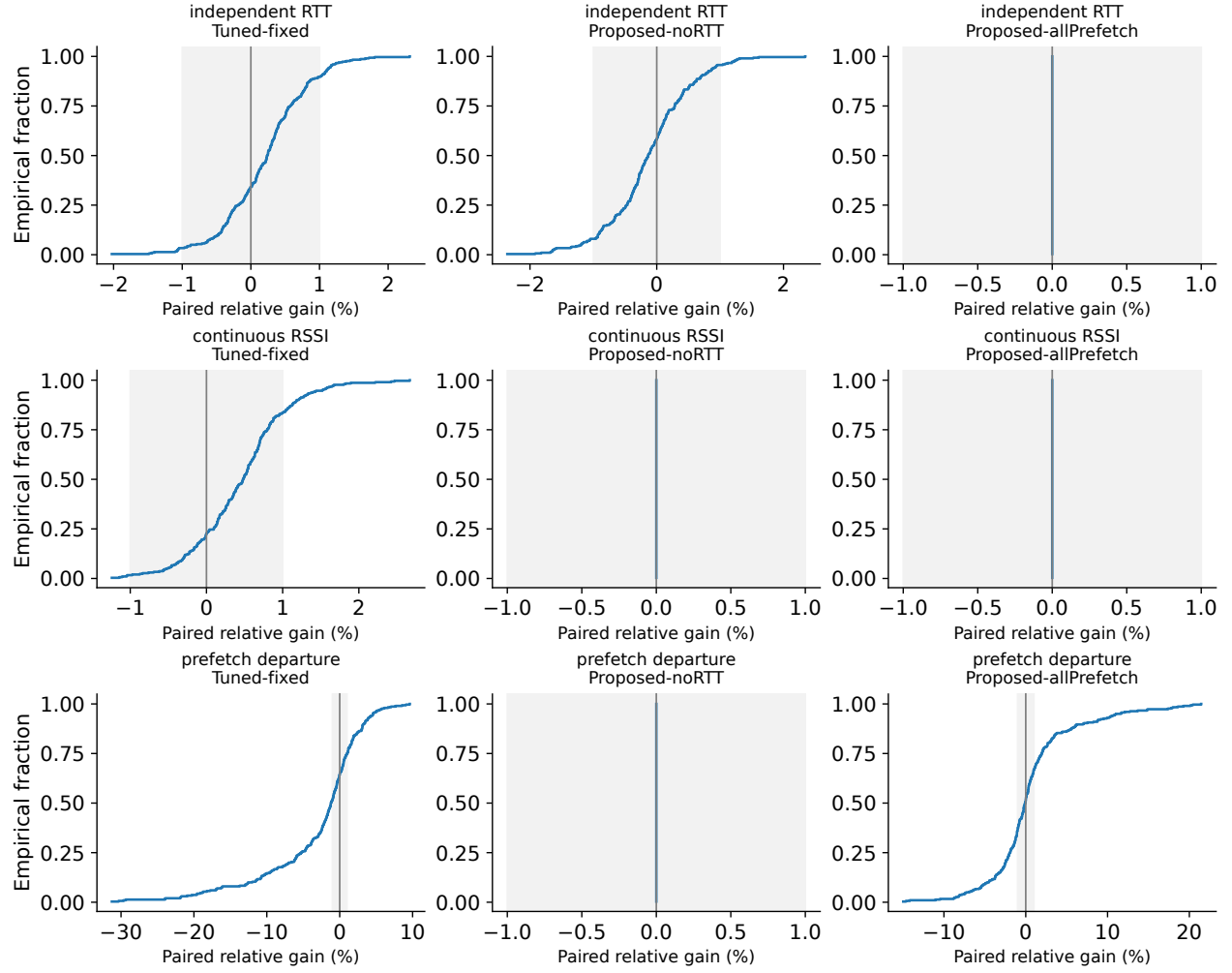
Cell	Candidate	Comparator	Mean gain (%)	Simultaneous CI (%)	Difference p	TOST p
dispatch 1000	Tuned-fixed	Fixed-one	0.00	[0.00, 0.00]	1.0000	< 0.0001
dispatch 20000	Tuned-fixed	Fixed-one	73.45	[73.34, 73.56]	< 0.0001	1.0000
dispatch 5000	Tuned-fixed	Fixed-one	2.58	[2.18, 2.97]	< 0.0001	1.0000

2 Scenario, Cost, and Calibration Settings

Cell	Type	Seeds	Service (ms)	Control (kB)
uniform	core	300	0.0	0.0
weak links	core	300	0.0	0.0
churn	core	300	0.0	0.0
combined	core	300	0.0	0.0
dispatch 0	sweep	200	0.0	64.0
dispatch 1	sweep	200	1.0	64.0
dispatch 10	sweep	200	10.0	64.0
dispatch 100	sweep	200	100.0	64.0
dispatch 1000	sweep	200	1000.0	64.0
dispatch 5000	sweep	200	5000.0	64.0
dispatch 20000	sweep	200	20000.0	64.0
independent RTT	activation	300	100.0	64.0
continuous RSSI	activation	300	100.0	64.0
prefetch departure	activation	300	100.0	64.0

Per-cell fixed-size candidates: 1,2,3,4,8,16,32. Calibration seeds: 5000–5019. All evaluation seeds are disjoint from calibration. The full machine-readable protocol and per-run counters are in `results/overhead_study/protocol.json` and `raw.json`.

3 Paired Distributions in Activation Cells



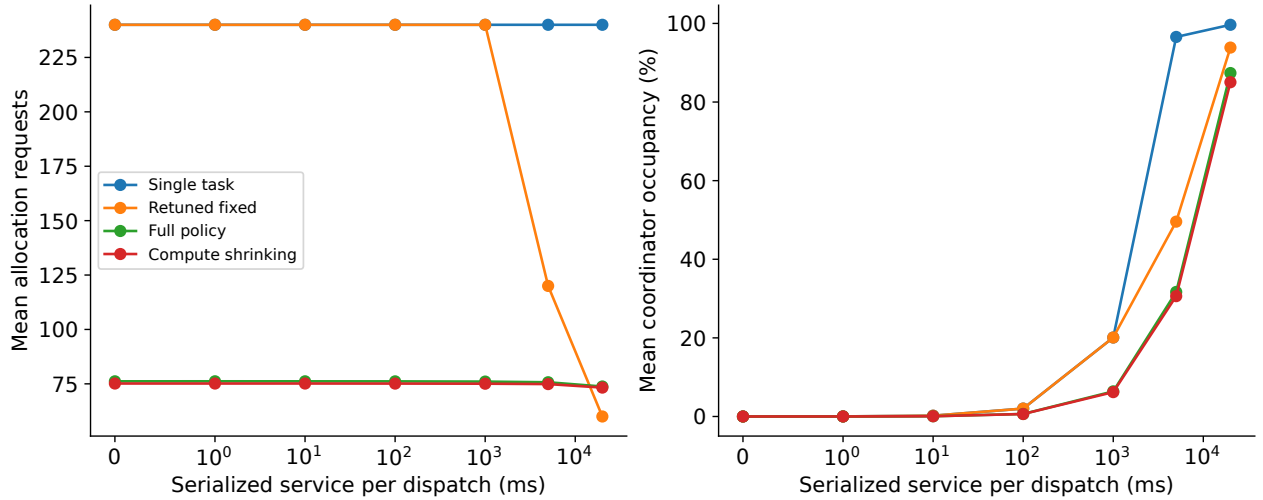
All 300 paired seeds are shown in each panel. Gray bands mark the declared equivalence margin; the distribution itself is descriptive.

4 Independent Tuning Grid



Each row uses 20 calibration worlds disjoint from evaluation. Entries are the percentage excess of calibration mean time over its row minimum. These are selection diagnostics, not significance tests or evaluation performance gains.

5 Coordinator Amortization Audit



Occupancy is completed service time divided by job completion time, averaged across seeds. This is modeled coordinator utilization, not a hardware measurement. The paired plots show whether allocation count and coordinator saturation explain the cost-sweep changes.

6 Resource Accounting

Cell	Policy	Dispatches	Prefetch	Dynamic PF	Data (MB)	Wasted compute
continuous RSSI	Compute-shrink	75.4	0.0	0.0	3964.3	0.0
continuous RSSI	Proposed	104.0	56.8	0.0	3966.1	0.0
continuous RSSI	Proposed-allPrefetch	104.0	56.8	0.0	3966.1	0.0
continuous RSSI	Proposed-noCutoff	104.0	88.9	0.0	3966.1	0.0
continuous RSSI	Proposed-noPrefetch	103.3	0.0	0.0	3966.1	0.0
continuous RSSI	Proposed-noRSSI	76.2	38.4	0.0	3964.3	0.0
continuous RSSI	Proposed-noRTT	104.0	56.8	0.0	3966.1	0.0
continuous RSSI	Tuned-fixed	240.0	0.0	0.0	3974.8	0.0
churn	Compute-shrink	71.2	0.0	0.0	4088.0	0.2
churn	Proposed	71.1	23.2	0.0	4087.9	0.2
churn	Proposed-allPrefetch	71.3	34.1	14.0	4088.0	0.2
churn	Proposed-noCutoff	71.6	41.6	0.0	4087.9	0.2
churn	Proposed-noPrefetch	71.2	0.0	0.0	4088.0	0.2
churn	Proposed-noRSSI	71.1	23.2	0.0	4087.9	0.2
churn	Proposed-noRTT	71.1	23.2	0.0	4087.9	0.2
churn	Tuned-fixed	242.0	0.0	0.0	3988.5	8.3
combined	Compute-shrink	75.2	0.0	0.0	4746.6	184.0
combined	Proposed	100.8	31.2	0.0	4323.2	81.7
combined	Proposed-allPrefetch	102.0	46.9	18.9	4337.0	83.5
combined	Proposed-noCutoff	100.9	49.0	0.0	4326.1	82.1
combined	Proposed-noPrefetch	100.1	0.0	0.0	4290.9	75.4
combined	Proposed-noRSSI	77.6	21.3	0.0	4782.2	168.0
combined	Proposed-noRTT	100.8	31.2	0.0	4323.2	81.7
combined	Tuned-fixed	248.0	0.0	0.0	4066.8	35.0
uniform	Compute-shrink	75.2	0.0	0.0	3958.0	0.0
uniform	Proposed	76.2	38.3	0.0	3958.0	0.0
uniform	Proposed-allPrefetch	76.2	38.3	0.0	3958.0	0.0
uniform	Proposed-noCutoff	76.6	65.9	0.0	3958.0	0.0
uniform	Proposed-noPrefetch	75.2	0.0	0.0	3958.0	0.0
uniform	Proposed-noRSSI	76.2	38.3	0.0	3958.0	0.0
uniform	Proposed-noRTT	76.2	38.3	0.0	3958.0	0.0
uniform	Tuned-fixed	240.0	0.0	0.0	3958.0	0.0
weak links	Compute-shrink	80.1	0.0	0.0	4617.1	153.1
weak links	Proposed	107.0	50.2	0.0	4230.2	66.1
weak links	Proposed-allPrefetch	107.0	50.2	0.0	4230.2	66.1
weak links	Proposed-noCutoff	107.0	76.7	0.0	4233.8	67.0
weak links	Proposed-noPrefetch	105.7	0.0	0.0	4193.1	62.2

Cell	Policy	Dispatches	Prefetch	Dynamic PF	Data (MB)	Wasted compute
weak links	Proposed-noRSSI	83.6	36.0	0.0	4669.4	143.1
weak links	Proposed-noRTT	107.0	50.2	0.0	4230.2	66.1
weak links	Tuned-fixed	247.1	0.0	0.0	4053.8	26.1
dispatch 0	Compute-shrink	75.2	0.0	0.0	3967.7	0.0
dispatch 0	Fixed-one	240.0	0.0	0.0	3978.2	0.0
dispatch 0	Proposed	76.2	38.7	0.0	3967.7	0.0
dispatch 0	Tuned-fixed	240.0	0.0	0.0	3978.2	0.0
dispatch 1	Compute-shrink	75.1	0.0	0.0	3967.7	0.0
dispatch 1	Fixed-one	240.0	0.0	0.0	3978.2	0.0
dispatch 1	Proposed	76.2	38.8	0.0	3967.7	0.0
dispatch 1	Tuned-fixed	240.0	0.0	0.0	3978.2	0.0
dispatch 10	Compute-shrink	75.2	0.0	0.0	3967.7	0.0
dispatch 10	Fixed-one	240.0	0.0	0.0	3978.2	0.0
dispatch 10	Proposed	76.2	38.7	0.0	3967.7	0.0
dispatch 10	Tuned-fixed	240.0	0.0	0.0	3978.2	0.0
dispatch 100	Compute-shrink	75.1	0.0	0.0	3967.7	0.0
dispatch 100	Fixed-one	240.0	0.0	0.0	3978.2	0.0
dispatch 100	Proposed	76.1	38.6	0.0	3967.7	0.0
dispatch 100	Tuned-fixed	240.0	0.0	0.0	3978.2	0.0
dispatch 1000	Compute-shrink	75.1	0.0	0.0	3967.7	0.0
dispatch 1000	Fixed-one	240.0	0.0	0.0	3978.2	0.0
dispatch 1000	Proposed	76.0	38.6	0.0	3967.7	0.0
dispatch 1000	Tuned-fixed	240.0	0.0	0.0	3978.2	0.0
dispatch 20000	Compute-shrink	73.3	0.0	0.0	3967.5	0.0
dispatch 20000	Fixed-one	240.0	0.0	0.0	3978.2	0.0
dispatch 20000	Proposed	73.8	35.4	0.0	3967.6	0.0
dispatch 20000	Tuned-fixed	60.0	0.0	0.0	3966.7	0.0
dispatch 5000	Compute-shrink	74.9	0.0	0.0	3967.6	0.0
dispatch 5000	Fixed-one	240.0	0.0	0.0	3978.2	0.0
dispatch 5000	Proposed	75.7	38.3	0.0	3967.7	0.0
dispatch 5000	Tuned-fixed	120.0	0.0	0.0	3970.5	0.0
independent RTT	Compute-shrink	75.4	0.0	0.0	3965.9	0.0
independent RTT	Proposed	169.6	106.7	0.0	3971.9	0.0
independent RTT	Proposed-allPrefetch	169.6	106.7	0.0	3971.9	0.0
independent RTT	Proposed-noCutoff	169.6	150.2	0.0	3971.9	0.0
independent RTT	Proposed-noPrefetch	166.7	0.0	0.0	3971.8	0.0
independent RTT	Proposed-noRSSI	169.6	106.7	0.0	3971.9	0.0

Cell	Policy	Dispatches	Prefetch	Dynamic PF	Data (MB)	Wasted compute
independent RTT	Proposed-noRTT	76.2	38.5	0.0	3966.0	0.0
independent RTT	Tuned-fixed	240.0	0.0	0.0	3976.4	0.0
prefetch departure	Compute-shrink	66.2	0.0	0.0	143.4	143.6
prefetch departure	Proposed	65.4	18.9	0.0	142.6	130.6
prefetch departure	Proposed-allPrefetch	67.5	38.2	17.5	148.2	238.2
prefetch departure	Proposed-noCutoff	65.5	29.9	0.0	142.6	130.6
prefetch departure	Proposed-noPrefetch	66.3	0.0	0.0	143.4	143.9
prefetch departure	Proposed-noRSSI	65.3	18.9	0.0	142.6	130.6
prefetch departure	Proposed-noRTT	65.4	18.9	0.0	142.6	130.6
prefetch departure	Tuned-fixed	122.0	0.0	0.0	141.7	39.3

Values are per-run means. Wasted compute is the reference-worker cost of computed tasks whose result is not accepted; it is not energy or wall-clock CPU time. Data includes actual fluid transfer volume, including control. Dispatches count requests, including ones canceled before service completes.

7 Reproducibility and Runtime Scope

The raw records retain world hashes, allocation hashes, completion, prefetch counts, dynamic prefetch counts, completed coordinator service time, control requested volume, actual data transmitted, unaccepted compute and expired assignments. Radio, coordinator and progress assumptions are separate. The corrected runtime cancels obsolete coordinator transfers for every policy. No retained-transfer contrast is presented as a published-system benchmark. Full scenario/policy switches, model hashes and every calibration and evaluation seed appear in `results/overhead_study/protocol.json`. Calibration curves and raw packet-level records are retained in `results/ns3` and `results/ns3_extended`.